

the creation and manipulation of ones and zeroes...

Visit www.pbs.org/transistor/album1/addl-bios/deforest.html for an amusing little video (it's at the right of the screen) of how the grid or gate that de Forest introduced helps control the flow of electrons. Visit www.pbs.org/wgbh/aso/databank/entries/btfore.html for what exactly the development consisted in. Also see figure below.



This was in everything electronic—before the transistor came in

Suffice it to say that it would have been impossible to make digital electronic computers without the introduction of the vacuum tube.

1919

It is difficult to explain to a general audience what a flip-flop is. It is also difficult to overstate its usefulness. Technically, it is a bi-stable multivibrator, and that means it has two stable states, meaning it can *hold* a bit. Sort of important when it comes to computers—a nice building block!

British physicist W H Eccles and F W Jordan published the first flip-flop circuit design in 1919. It was then called the Eccles-Jordan circuit.

A “stable state” means just that—there are electronic circuits or devices that can go into one state or another when propelled by a current, but without staying in that state. A bi-stable multivibrator, then, is important because it has *two* stable states. You give it a current and tell it to become a 1, and it obeys; change it and tell it to become a zero, and it does so.

Now here's from foldoc.org: “Early literature refers to the ‘Eccles-Jordan circuit’ and the ‘Eccles-Jordan binary counter,’